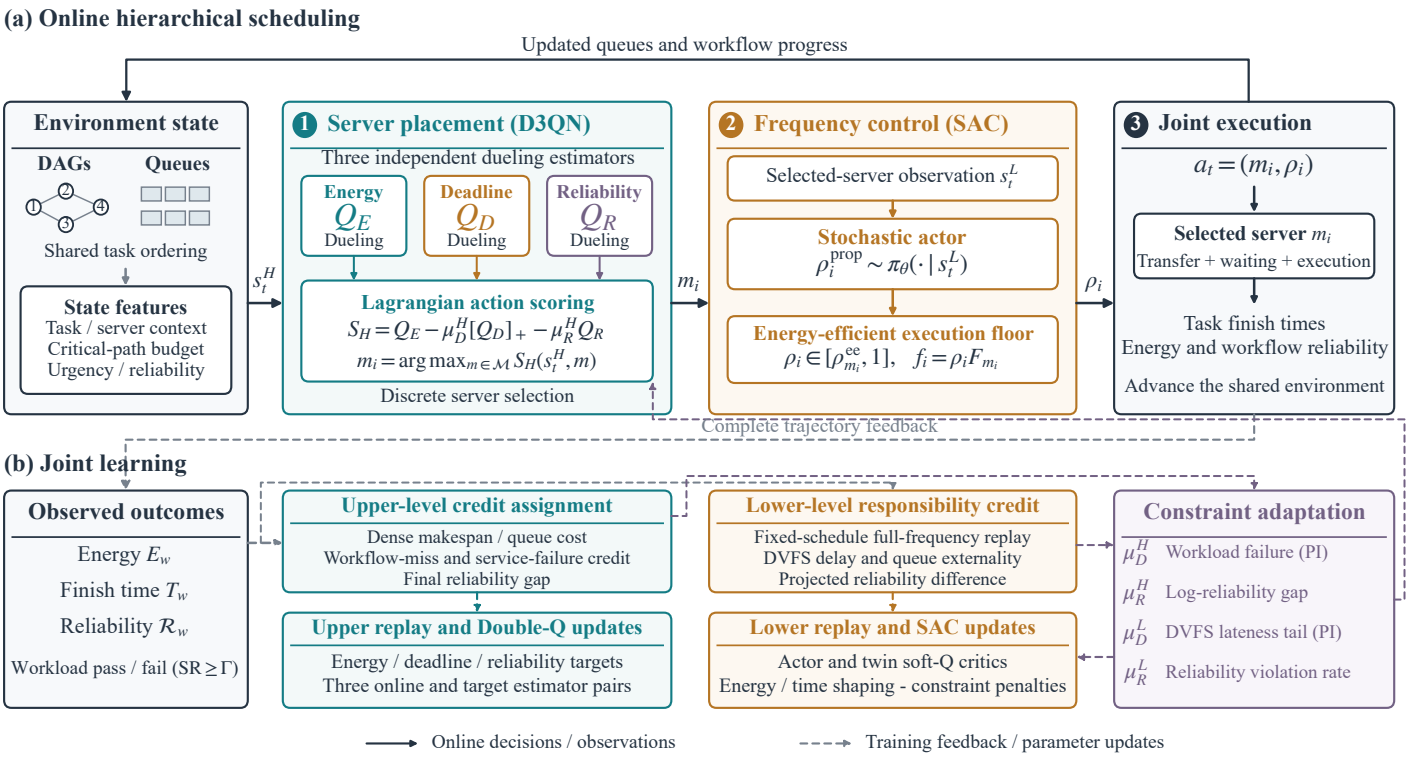

(a) Online hierarchical scheduling
Updated queues and workflow progress
Environment state
DAGs
Queues
Shared task ordering
State features
Task / server context
Critical-path budget
Urgency / reliability
s_t^H
1 Server placement (D3QN)
Three independent dueling estimators
Energy Q_E Dueling
Deadline Q_D Dueling
Reliability Q_R Dueling
Lagrangian action scoring
$S_H = Q_E - \mu_D^H [Q_D]_+ - \mu_R^H Q_R$
$m_i = \arg\max_{m \in \mathcal{M}} S_H(s_t^H, m)$
Discrete server selection
m_i
2 Frequency control (SAC)
Selected-server observation s_t^L
Stochastic actor
$\rho_i^{\text{prop}} \sim \pi_\theta(\cdot \mid s_t^L)$
Energy-efficient execution floor
$\rho_i \in [\rho_{m_i}^{\text{ee}}, 1], \quad f_i = \rho_i F_{m_i}$
ρ_i
3 Joint execution
$a_t = (m_i, \rho_i)$
Selected server m_i
Transfer + waiting + execution
Task finish times
Energy and workflow reliability
Advance the shared environment
Complete trajectory feedback
(b) Joint learning
Observed outcomes
Energy E_w
Finish time T_w
Reliability $\mathcal{R}_w$
Workload pass / fail (SR $\geq \Gamma$)
Upper-level credit assignment
Dense makespan / queue cost
Workflow-miss and service-failure credit
Final reliability gap
Upper replay and Double-Q updates
Energy / deadline / reliability targets
Three online and target estimator pairs
Lower-level responsibility credit
Fixed-schedule full-frequency replay
DVFS delay and queue externality
Projected reliability difference
Lower replay and SAC updates
Actor and twin soft-Q critics
Energy / time shaping - constraint penalties
Constraint adaptation
μ_D^H Workload failure (PI)
μ_R^H Log-reliability gap
μ_D^L DVFS lateness tail (PI)
μ_R^L Reliability violation rate
Online decisions / observations
Training feedback / parameter updates